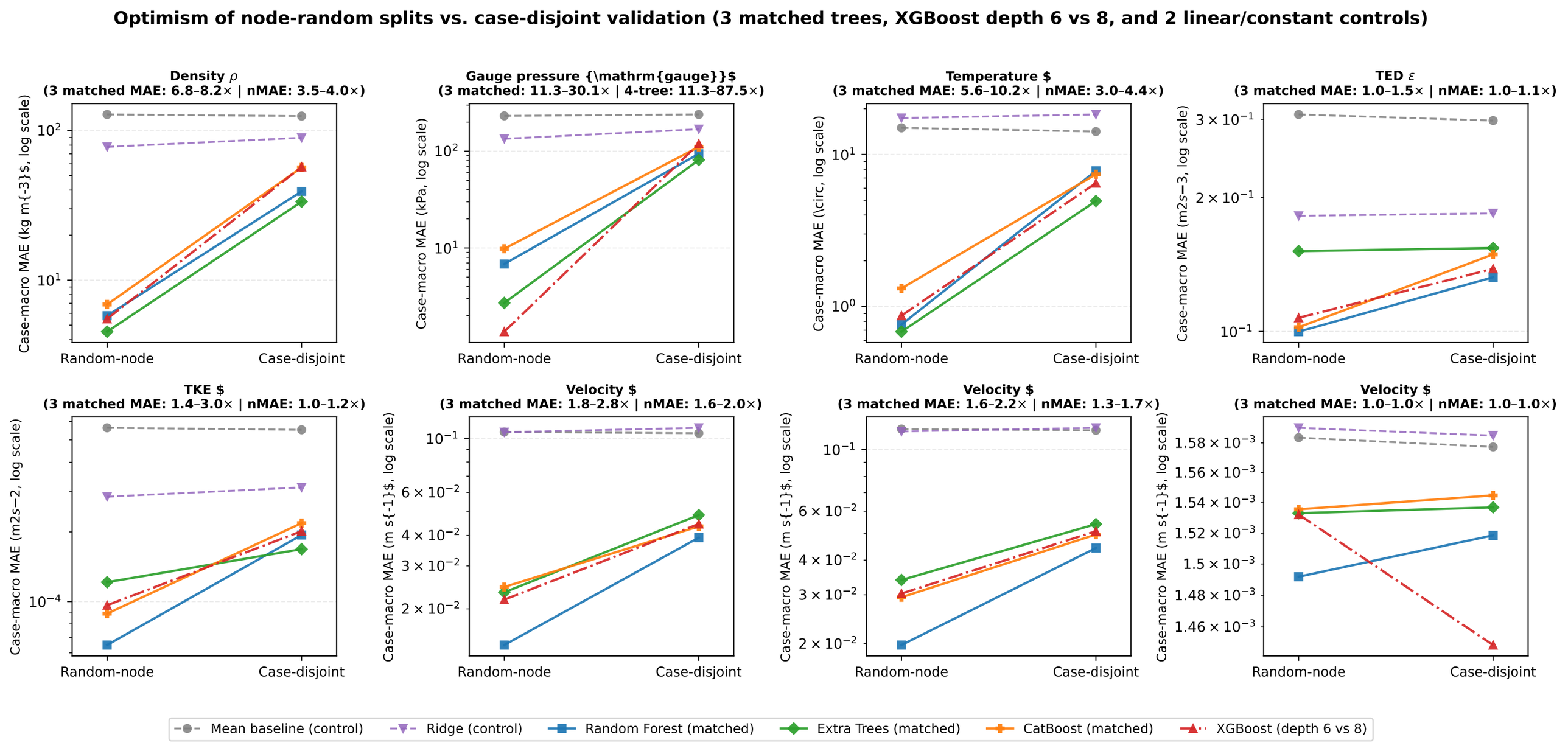
Optimism of node-random splits vs. case-disjoint validation (3 matched trees, XGBoost depth 6 vs 8, and 2 linear/constant controls)
Density ρ
(3 matched MAE: 6.8–8.2× | nMAE: 3.5–4.0×)
Case-macro MAE (kg m{-3}$, log scale)
Gauge pressure {\mathrm{gauge}}$
(3 matched: 11.3–30.1× | 4-tree: 11.3–87.5×)
Case-macro MAE (kPa, log scale)
Temperature $
(3 matched MAE: 5.6–10.2× | nMAE: 3.0–4.4×)
Case-macro MAE (\circ, log scale)
TED ε
(3 matched MAE: 1.0–1.5× | nMAE: 1.0–1.1×)
Case-macro MAE (m2s−3, log scale)
TKE $
(3 matched MAE: 1.4–3.0× | nMAE: 1.0–1.2×)
Case-macro MAE (m2s−2, log scale)
Velocity $
(3 matched MAE: 1.8–2.8× | nMAE: 1.6–2.0×)
Case-macro MAE (m s{-1}$, log scale)
Velocity $
(3 matched MAE: 1.6–2.2× | nMAE: 1.3–1.7×)
Case-macro MAE (m s{-1}$, log scale)
Velocity $
(3 matched MAE: 1.0–1.0× | nMAE: 1.0–1.0×)
Case-macro MAE (m s{-1}$, log scale)
Random-node
Case-disjoint
Mean baseline (control)
Ridge (control)
Random Forest (matched)
Extra Trees (matched)
CatBoost (matched)
XGBoost (depth 6 vs 8)